12TH CHEMISTRY · CHAPTER 17 · FULL CHAPTER MCQS

Group 2 Elements

84 topic-wise MCQs from the complete chapter with answers and reasons

17.1 PROPERTIES AND TRENDS OF GROUP 2 ELEMENTS

1. Group 2 elements are also called:

(a) Alkali metals
(b) Transition metals
(c) Halogens
(d) Alkaline earth metals

Answer: (d) Alkaline earth metals — Due to their metallic nature and occurrence in the earth's crust.

2. Group 2 elements belong to which block?

(a) p-block
(b) s-block
(c) d-block
(d) f-block

Answer: (b) s-block — Their valence electrons are in an s-subshell.

3. The general valence shell configuration of Group 2 elements is:

(a) ns^2
(b) ns^1
(c) ns^2np^1
(d) ns^2np^2

Answer: (a) ns^2 — Two electrons in the outermost s-subshell.

4. The electron configuration of calcium is:

(a) [Ne] $3s^2$
(b) [Kr] $5s^2$
(c) [Ar] $4s^2$
(d) [He] $2s^2$

Answer: (c) [Ar] $4s^2$ — Atomic number 20.

5. Which Group 2 element has the highest first ionization energy?

(a) Be
(b) Mg
(c) Ca
(d) Ba

Answer: (a) Be — Be is the smallest atom (900).

6. Which element has the highest melting point in Group 2?

(a) Mg
(b) Be
(c) Ca
(d) Ba

Answer: (b) Be — 1287 °C.

7. Which element has the lowest melting point among Be–Ba?

(a) Be
(b) Sr
(c) Mg
(d) Ba

Answer: (c) Mg — 650 °C.

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8. The Group 2 element with the lowest density is:

(a) Be (b) Ba
(c) Mg (d) Ca

Answer: (d) Ca — 1.54 g/cm^3.

9. Down Group 2, the metallic radius:

(a) Increases (b) Decreases
(c) Stays constant (d) First rises then falls

Answer: (a) Increases — 112 pm (Be) to 222 pm (Ba).

10. The ionic radius of Be^{2+} is:

(a) 65 pm (b) 31 pm
(c) 99 pm (d) 135 pm

Answer: (b) 31 pm — The smallest ion of the group.

11. Electronegativity down Group 2:

(a) Increases (b) Is zero
(c) Remains 1.5 (d) Decreases

Answer: (d) Decreases — 1.5 (Be) to 0.9 (Ba).

12. The most negative standard electrode potential belongs to:

(a) Be (b) Mg
(c) Ba (d) Ca

Answer: (c) Ba — –2.91 V.

13. Group 2 elements form ions with a charge of:

(a) +1 (b) +2
(c) +3 (d) –2

Answer: (b) +2 — They lose two ns electrons.

14. The correct reactivity series is:

(a) Be < Mg < Ca < Sr < Ba
(b) Ba < Sr < Ca < Mg < Be
(c) Mg < Be < Ca < Ba < Sr
(d) Be < Ca < Mg < Sr < Ba

Answer: (a) Be < Mg < Ca < Sr < Ba — Reactivity increases down the group.

15. Reactivity increases down Group 2 because:

(a) Ionization energy increases (b) Atomic size decreases
(c) Ionization energy decreases (d) Shielding decreases

Answer: (c) Ionization energy decreases — Electrons are removed more easily.

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16. Which element is an important part of chlorophyll?

(a) Ca (b) Ba
(c) Be (d) Mg

Answer: (d) Mg — Mg is the central atom of chlorophyll.

17. Which metal makes copper alloys harder for aerospace materials?

(a) Mg (b) Be
(c) Sr (d) Ca

Answer: (b) Be — Stated in the introduction.

17.2 CHEMICAL REACTIONS OF GROUP 2 ELEMENTS

18. The general formula of Group 2 oxides is:

(a) M_2O (b) MO
(c) MO_2 (d) M_2O_3

Answer: (b) MO — $2M + O_2 \rightarrow 2MO$.

19. Beryllium reacts with oxygen only above:

(a) 100 °C (b) 300 °C
(c) 600 °C (d) 1287 °C

Answer: (c) 600 °C — It is protected by an oxide layer.

20. Calcium burns in oxygen with a flame of which colour?

(a) Brick red (b) Green
(c) White (d) Blue

Answer: (a) Brick red — Stated in section 17.2.1.

21. Group 2 oxides are generally:

(a) Acidic (b) Coloured liquids
(c) Neutral (d) Basic

Answer: (d) Basic — White basic solids.

22. The oxide ion reacts with water to form:

(a) OH^- ions (b) H^+ ions
(c) O_2 gas (d) H_2 gas

Answer: (a) OH^- ions — $O^{2-} + H_2O \rightarrow 2OH^-$.

23. Beryllium does not react with water because of:

(a) Its large size (b) Its low density
(c) Its low melting point (d) Its protective oxide layer

Answer: (d) Its protective oxide layer — Even on heating.

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24. Magnesium with cold water forms:

(a) MgO and H_2
(b) $Mg(OH)_2$ and H_2
(c) MgO only
(d) MgH_2

Answer: (b) $Mg(OH)_2$ and H_2 — A very slow reaction.

25. Magnesium with steam forms:

(a) $Mg(OH)_2$ and H_2
(b) MgO and O_2
(c) MgO and H_2
(d) $Mg(OH)_2$ only

Answer: (c) MgO and H_2 — Faster than with cold water.

26. Calcium hydroxide in water is:

(a) Very soluble
(b) A gas
(c) Completely insoluble
(d) Sparingly soluble

Answer: (d) Sparingly soluble — It forms a white precipitate.

27. Group 2 metals react with dilute HCl to give:

(a) Salt and water
(b) Oxide and hydrogen
(c) Salt and hydrogen
(d) Hydroxide and chlorine

Answer: (c) Salt and hydrogen — $M + 2HCl \rightarrow MCl_2 + H_2$.

28. The reaction of Ca with H_2SO_4 stops early because $CaSO_4$ is:

(a) Very soluble
(b) Sparingly soluble
(c) A gas
(d) Coloured

Answer: (b) Sparingly soluble — It coats the metal surface.

29. The basic character of Group 2 hydroxides down the group:

(a) Increases
(b) Decreases
(c) Is constant
(d) Becomes acidic

Answer: (a) Increases — Due to increasing solubility.

17.3 COMPARISON WITH GROUP 1 ELEMENTS

30. Group 1 metals form ions of the type:

(a) M^{2+}
(b) M^{3+}
(c) M^{+}
(d) M^{-}

Answer: (c) M^{+} — They have one valence electron.

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31. Compared with Group 1 metals of the same period, Group 2 metals are:

(a) More reactive (b) Less reactive
(c) Equally reactive (d) Unreactive

Answer: (b) Less reactive — Higher ionization energies.

32. Which can be cut with a knife?

(a) Group 2 metals (b) Neither
(c) Both (d) Group 1 metals

Answer: (d) Group 1 metals — They are very soft.

33. The general formula of a Group 1 oxide is:

(a) M_2O (b) MO
(c) MO_2 (d) M_3O

Answer: (a) M_2O — e.g., Na_2O.

34. Melting points of Group 2 metals compared with Group 1 are:

(a) Lower (b) Equal
(c) Higher (d) Zero

Answer: (c) Higher — Table 17.2.

35. Group 1 hydroxides are:

(a) Insoluble weak bases (b) Highly soluble strong bases
(c) Acidic (d) Sparingly soluble

Answer: (b) Highly soluble strong bases — Table 17.2.

17.4 THERMAL DECOMPOSITION OF CARBONATES AND NITRATES

36. Thermal decomposition is usually:

(a) Exothermic (b) Explosive
(c) Neither (d) Endothermic

Answer: (d) Endothermic — Energy is needed to break bonds.

37. Group 2 carbonates decompose into:

(a) Metal and CO_2
(b) Metal oxide and CO_2
(c) Metal oxide and CO
(d) Metal hydroxide and CO_2

Answer: (b) Metal oxide and CO_2 — $MCO_3 \rightarrow MO + CO_2$.

38. The least thermally stable carbonate is:

(a) $BaCO_3$ (b) $CaCO_3$
(c) $BeCO_3$ (d) $SrCO_3$

Answer: (c) $BeCO_3$ — It decomposes at 250 °C.

39. $CaCO_3$ decomposes at 1 atm at about:

(a) 900 °C (b) 540 °C
(c) 250 °C (d) 1360 °C

Answer: (a) 900 °C — Table 17.3.

40. The decomposition temperature of $BaCO_3$ is:

(a) 1289 °C (b) 900 °C
(c) 1360 °C (d) 540 °C

Answer: (c) 1360 °C — The highest of the group.

41. Polarizing power is greatest for:

(a) Be^{2+} (b) Ba^{2+}
(c) Sr^{2+} (d) Ca^{2+}

Answer: (a) Be^{2+} — The smallest cation.

42. The brown gas given off when a Group 2 nitrate is heated is:

(a) O_2 (b) CO_2
(c) N_2 (d) NO_2

Answer: (d) NO_2 — Nitrogen dioxide.

43. In $2Ca(NO_3)_2 \rightarrow 2CaO + xNO_2 + O_2$, x is:

(a) 2 (b) 4
(c) 1 (d) 3

Answer: (b) 4 — Balanced equation.

44. The most thermally stable nitrate is:

(a) $Ba(NO_3)_2$
(b) $Be(NO_3)_2$
(c) $Mg(NO_3)_2$
(d) $Ca(NO_3)_2$

Answer: (a) $Ba(NO_3)_2$ — Ba^{2+} has the lowest polarizing power.

45. A small cation makes a carbonate less stable because it:

(a) Strengthens the C–O bonds (b) Lowers the charge
(c) Distorts the carbonate ion (d) Adds electrons

Answer: (c) Distorts the carbonate ion — Distortion weakens the bonds.

17.5 SOLUBILITY OF SULPHATES AND HYDROXIDES

46. Enthalpy of hydration is always:

(a) Endothermic (b) Positive
(c) Zero (d) Exothermic

Answer: (d) Exothermic — Energy is released.

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47. Lattice enthalpy, as defined in the textbook, is always:

(a) Exothermic (b) Endothermic
(c) Zero (d) Negative

Answer: (b) Endothermic — Energy needed to break the lattice.

48. Down Group 2, ΔH_{hyd} of M^{2+} becomes:

(a) More negative (b) Positive
(c) Less negative (d) Zero

Answer: (c) Less negative — The ions get larger.

49. The solubility of Group 2 hydroxides down the group:

(a) Increases (b) Decreases
(c) Is constant (d) Is zero

Answer: (a) Increases — $Mg(OH)_2$ to $Ba(OH)_2$.

50. The solubility of Group 2 sulphates down the group:

(a) Increases (b) Is constant
(c) Decreases (d) First rises

Answer: (c) Decreases — $MgSO_4$ soluble, $BaSO_4$ insoluble.

51. The most soluble hydroxide is:

(a) $Ba(OH)_2$ (b) $Mg(OH)_2$
(c) $Ca(OH)_2$ (d) $Sr(OH)_2$

Answer: (a) $Ba(OH)_2$ — 1.5×10^{-2} mol/100 g.

52. The most soluble sulphate is:

(a) $BaSO_4$ (b) $CaSO_4$
(c) $SrSO_4$ (d) $MgSO_4$

Answer: (d) $MgSO_4$ — 1.83×10^{-1} mol/100 g.

53. For hydroxides, which decreases more down the group?

(a) Hydration enthalpy (b) Lattice enthalpy
(c) Ionic charge (d) Atomic number

Answer: (b) Lattice enthalpy — So solubility increases.

54. For sulphates, which decreases more down the group?

(a) Hydration enthalpy (b) Lattice enthalpy
(c) Ionic charge (d) Anion size

Answer: (a) Hydration enthalpy — So solubility decreases.

55. Higher solubility is generally favoured by a ΔH_{soln} that is:

(a) More positive (b) Zero
(c) More negative (d) Unchanged

Answer: (c) More negative — Stated in section 17.5.

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17.6 COMPLEX ION FORMATION

56. In a complex ion the central metal ion acts as a:

(a) Lewis acid (b) Lewis base
(c) Reducing agent (d) Catalyst

Answer: (a) Lewis acid — Electron pair acceptor.

57. The number of coordinate bonds to the central metal ion is the:

(a) Oxidation number (b) Atomic number
(c) Coordination number (d) Mass number

Answer: (c) Coordination number — Definition.

58. The coordination number of Be in its hydrated ion is:

(a) 6 (b) 8
(c) 2 (d) 4

Answer: (d) 4 — $[Be(H_2O)_4]^{2+}$.

59. The hydrated magnesium ion is:

(a) $[Mg(H_2O)_4]^{2+}$
(b) $[Mg(H_2O)_6]^{2+}$
(c) $[Mg(H_2O)_2]^{2+}$
(d) $[Mg(H_2O)_8]^{2+}$

Answer: (b) $[Mg(H_2O)_6]^{2+}$ — Coordination number 6.

60. EDTA donates how many electron pairs?

(a) Two (b) Four
(c) Six (d) One

Answer: (c) Six — A six electron pair donor.

61. The charge on the calcium–EDTA complex is:

(a) 2+ (b) Zero
(c) 4– (d) 2–

Answer: (d) 2– — $[Ca(EDTA)]^{2-}$.

62. EDTA is used in:

(a) Water softening (b) Fireworks
(c) Cement making (d) Electrolysis

Answer: (a) Water softening — Removes Ca^{2+} and Mg^{2+}.

63. Which of these is NOT listed as a Lewis base in the chapter?

(a) NH_3 (b) Mg^{2+}
(c) H_2O (d) Cl^-

Answer: (b) Mg^{2+} — Metal ions are Lewis acids.

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17.7 EXTRACTION AND PURIFICATION

64. The Dow process extracts:

(a) Magnesium (b) Calcium
(c) Barium (d) Beryllium

Answer: (a) Magnesium — From seawater.

65. In the Dow process, seawater is first treated with:

(a) HCl (b) $CaCO_3$
(c) NaCl (d) $Ca(OH)_2$

Answer: (d) $Ca(OH)_2$ — It precipitates $Mg(OH)_2$.

66. $Mg(OH)_2$ is converted into $MgCl_2$ with:

(a) Chlorine (b) Hydrochloric acid
(c) Sodium chloride (d) Water

Answer: (b) Hydrochloric acid — $Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$.

67. In the electrolysis of molten $MgCl_2$, magnesium forms at the:

(a) Anode (b) Salt bridge
(c) Cathode (d) Surface

Answer: (c) Cathode — Chlorine forms at the anode.

68. Quicklime is:

(a) $CaCO_3$ (b) $CaSO_4$
(c) $Ca(OH)_2$ (d) CaO

Answer: (d) CaO — Calcium oxide.

69. The temperature in a lime kiln is typically:

(a) 900–1000 °C (b) 250–300 °C
(c) 100 °C (d) 2000 °C

Answer: (a) 900–1000 °C — Section 17.7.2.

70. Heating limestone strongly in a kiln is called:

(a) Hydration (b) Calcination
(c) Electrolysis (d) Scrubbing

Answer: (b) Calcination — Thermal decomposition.

71. Removing CO_2 from the kiln shifts the equilibrium:

(a) To the left (b) Nowhere
(c) To the right (d) To the reactants

Answer: (c) To the right — More CaO is formed.

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17.8 USES OF GROUP 2 COMPOUNDS

72. Gypsum is:

(a) $CaSO_4 \cdot 2H_2O$
(b) $CaSO_4 \cdot ½H_2O$
(c) $MgSO_4 \cdot 7H_2O$
(d) $CaCO_3$

Answer: (a) $CaSO_4 \cdot 2H_2O$ — Used to make plasterboard.

73. Plaster of Paris is:

(a) $CaSO_4 \cdot 2H_2O$
(b) $BaSO_4$
(c) CaO
(d) $CaSO_4 \cdot ½H_2O$

Answer: (d) $CaSO_4 \cdot ½H_2O$ — Section 17.8.1.

74. Slaked lime is:

(a) CaO
(b) $CaCO_3$
(c) $Ca(OH)_2$
(d) MgO

Answer: (c) $Ca(OH)_2$ — Calcium hydroxide.

75. Flue gas desulphurization removes:

(a) CO_2
(b) SO_2
(c) NO_2
(d) O_2

Answer: (b) SO_2 — Prevents acid rain.

76. MgO is used for lining furnaces because of its:

(a) Low density
(b) Colour
(c) Solubility
(d) High melting point

Answer: (d) High melting point — A refractory material.

77. The red colour in fireworks is produced by compounds of:

(a) Strontium
(b) Barium
(c) Beryllium
(d) Magnesium

Answer: (a) Strontium — "Did You Know?" box.

78. Beryllium is used in X-ray tubes because it is:

(a) Opaque to X-rays
(b) Radioactive
(c) Transparent to X-rays
(d) Magnetic

Answer: (c) Transparent to X-rays — "Did You Know?" box.

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79. "Milk of Magnesia" is:

(a) $MgSO_4$ (b) $Mg(OH)_2$
(c) MgO (d) $MgCO_3$

Answer: (b) $Mg(OH)_2$ — An antacid and mild laxative.

80. Epsom salt is:

(a) $MgSO_4 \cdot 7H_2O$
(b) $CaSO_4 \cdot 2H_2O$
(c) $BaSO_4$
(d) $MgCl_2$

Answer: (a) $MgSO_4 \cdot 7H_2O$ — Used as a laxative and in baths.

81. The compound used in "barium meals" is:

(a) $BaCO_3$ (b) $BaCl_2$
(c) $BaSO_4$ (d) $Ba(OH)_2$

Answer: (c) $BaSO_4$ — Opaque to X-rays and insoluble.

82. Calcium supplements help to prevent:

(a) Anaemia (b) Osteoporosis
(c) Goitre (d) Diabetes

Answer: (b) Osteoporosis — Strong bones and teeth.

83. Lime is added to soil in order to:

(a) Lower the pH (b) Kill plants
(c) Add nitrogen only (d) Raise the pH

Answer: (d) Raise the pH — It neutralises acidity.

84. In plants, calcium is crucial for:

(a) Chlorophyll (b) Cell wall development
(c) Nitrogen fixation (d) Flower colour

Answer: (b) Cell wall development — Section 17.8.3.

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